



Experiment 7

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1. Aim:

To implement JPA entity mapping, advanced queries, fetch strategies, and database performance optimization using Spring Boot, JPA, and Hibernate.

2. Objective: After completing this experiment, the student will be able to:

- To create JPA entity classes and map them to database tables using annotations.
- To implement One-to-Many relationships between entities using JPA.
- To perform advanced database queries using JPQL and pagination techniques.
- To analyze Lazy and Eager fetch strategies and their impact on performance.
- To optimize database performance using indexing and query tuning.

3. Implementation/Code:

Appointment.java:

```
package com.example.demo.entity;
```

```
import jakarta.persistence.*;  
import lombok.*;  
import java.time.LocalDateTime;  
import java.util.HashSet;  
import java.util.Set;
```

```
/**
```

```
 * Appointment Entity - Represents an appointment between Patient and Doctor  
 * Demonstrates Many-to-One relationships and Eager fetching for core entities  
 * Demonstrates N+1 problem when querying without proper fetch strategy  
 */
```

```
@Entity
```

```
@Table(name = "appointment", indexes = {
```

```
    @Index(name = "idx_appointment_date", columnList = "appointment_date"),
```



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```
@Index(name = "idx_patient_id", columnList = "patient_id"),
@Index(name = "idx_doctor_id", columnList = "doctor_id"),
@Index(name = "idx_appointment_status", columnList = "status")
})
@Data
@NoArgsConstructor
@AllArgsConstructor
@Builder
public class Appointment {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    /**
     * Many-to-One with Patient
     * Using EAGER fetching for accessing patient directly helps avoid lazy loading issues
     * However, when fetching appointments, this can lead to inefficiency
     */
    @ManyToOne(fetch = FetchType.EAGER)
    @JoinColumn(name = "patient_id", nullable = false)
    private Patient patient;

    /**
     * Many-to-One with Doctor
     * EAGER fetching ensures doctor details are loaded with appointment
     * This can demonstrate the performance trade-offs
     */
    @ManyToOne(fetch = FetchType.EAGER)
    @JoinColumn(name = "doctor_id", nullable = false)
    private Doctor doctor;

    @Column(name = "appointment_date", nullable = false)
    private LocalDateTime appointmentDate;

    @Column(name = "duration_minutes")
    private Integer durationMinutes;

    @Column(length = 1000)
    private String reason;
```



```
@Column(length = 2000)
```

```
private String notes;
```

```
@Enumerated(EnumType.STRING)
```

```
@Column(nullable = false)
```

```
private AppointmentStatus status;
```

```
@Column(name = "created_at")
```

```
private LocalDateTime createdAt;
```

```
@Column(name = "updated_at")
```

```
private LocalDateTime updatedAt;
```

```
/**
```

```
 * One-to-Many with Prescription
```

```
 * LAZY loading ensures prescriptions are only loaded when needed
```

```
 */
```

```
@OneToMany(mappedBy = "appointment", fetch = FetchType.LAZY, cascade =  
CascadeType.ALL, orphanRemoval = true)
```

```
@Builder.Default
```

```
private Set<Prescription> prescriptions = new HashSet<>();
```

```
@PrePersist
```

```
protected void onCreate() {
```

```
    createdAt = LocalDateTime.now();
```

```
    updatedAt = LocalDateTime.now();
```

```
    status = AppointmentStatus.SCHEDULED;
```

```
}
```

```
@PreUpdate
```

```
protected void onUpdate() {
```

```
    updatedAt = LocalDateTime.now();
```

```
}
```

```
public enum AppointmentStatus {
```

```
    SCHEDULED, COMPLETED, CANCELLED, RESCHEDULED, NO_SHOW
```

```
}
```

```
}
```



4. Output:

```
/* insert
for
    com.example.demo.entity.Prescription
*/insert
into
    prescription (appointment_id, created_at, dosage, duration_days, expiry_date, frequency, instructions, medication_name, patient_id, prescribed_date, id)
values
    (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, default)
Hibernate:
/* insert
for
    com.example.demo.entity.Prescription
*/insert
into
    prescription (appointment_id, created_at, dosage, duration_days, expiry_date, frequency, instructions, medication_name, patient_id, prescribed_date, id)
values
    (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, default)
2026-03-16T12:01:53.983+05:30 INFO 18276 --- [demo] [      main] com.example.demo.DataInitializer      : ? Created 3 patients
2026-03-16T12:01:53.984+05:30 INFO 18276 --- [demo] [      main] com.example.demo.DataInitializer      : ? Created 3 doctors
2026-03-16T12:01:53.985+05:30 INFO 18276 --- [demo] [      main] com.example.demo.DataInitializer      : ? Created 3 appointments
2026-03-16T12:01:53.986+05:30 INFO 18276 --- [demo] [      main] com.example.demo.DataInitializer      : ? Created 3 prescriptions
2026-03-16T12:01:53.987+05:30 INFO 18276 --- [demo] [      main] com.example.demo.DataInitializer      : ===== Sample Data Initialization Complete =====
2026-03-16T12:01:56.017+05:30 INFO 18276 --- [demo] [on(2)-127.0.0.1] o.a.c.c.c.[Tomcat].[localhost].[/]    : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-03-16T12:01:56.018+05:30 INFO 18276 --- [demo] [on(2)-127.0.0.1] o.s.web.servlet.DispatcherServlet     : Initializing Servlet 'dispatcherServlet'
2026-03-16T12:01:56.020+05:30 INFO 18276 --- [demo] [on(2)-127.0.0.1] o.s.web.servlet.DispatcherServlet     : Completed initialization in 2 ms
```

5. Learning Outcome:

- Understand how JPA maps Java objects to relational database tables.
- Develop efficient database queries using JPQL.
- Apply pagination techniques for handling large datasets.
- Evaluate the performance impact of different fetch strategies.
- Optimize backend applications using indexing and query optimization techniques.